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Calibration scheme for Edgar's pXRF data

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1 Overview

Elemental concentrations of predominantly shale and carbonate samples from the Isa Superbasin were measured at the Research School of Earth Sciences, Australian National University, using a portable X-ray fluorescence (pXRF) spectrometer (Olympus Vanta, instrument serial no. 842861) operated in *Geochem (3-Beam)* mode with a nominal measurement time of 40 s per beam for all standard and sample analyses. Although the instrument applies an internal standardisation and reports concentrations in parts per million (ppm), the raw output is known to still depart systematically from certified reference values, with offsets and scatter that can vary between analytical sessions due to instrument drift, replacement of pXRF windows, ambient temperature changes, differences in placement geometry of samples on the analyser, etc. We therefore developed a session-resolved calibration procedure, described below, to convert raw pXRF concentrations to values that are comparable with high-accuracy reference compositions.

1.1 Sources of error in pXRF data

Although it might seem esoteric, developing a working procedure that I was fully comfortable with involved diving deep into the different types of error that can arise in pXRF analysis and the methodology has iteratively evolved as I have better understood them. It's worth having a quick overview before we begin as it will help explain some of the methodological intricacies!

A pXRF spectrometer determines elemental concentrations by irradiating a sample with a primary X-ray beam generated by a miniaturised X-ray tube. When these primary X-rays strike the sample, they eject inner-shell electrons from atoms in the material, causing outer-shell electrons to drop inward and release energy as secondary (fluorescent) X-rays with wavelengths characteristic of each element present. A solid-state detector (typically a silicon drift detector) records the energies and count rates of these fluorescent X-rays, producing an energy-dispersive spectrum that is deconvolved by the instrument software to identify elements and estimate their concentrations. Since fluorescence signal depends not only on the abundance of the element being analysed, but also on the overall composition of the sample matrix (which governs how efficiently primary X-rays penetrate the sample and how efficiently secondary X-rays escape to the detector), the raw instrument output requires calibration against reference materials of known composition. In typical usage, calibration is performed empirically using a suite of geochemical reference standards measured under the same conditions as the unknowns, with the calibration relating the instrument-reported concentration to the certified reference value via a linear scaling factor.

Errors that are likely to be random between different analyses of the same session arise from counting statistics (the Poisson noise of the detected photon count, which is formally reported by the instrument), slight variability of the standards between replicate measurements (reference puck heterogeneity and/or positioning geometry effects), and any short-term electrical noise. Partially to combat these random errors, the standards are typically measured repeatedly throughout the session, such that their average stabilises.

Errors that may vary systematically from sample to sample across the course of a session or between sessions include longer-term instrument drift (e.g., detector ageing, X-ray tube intensity changes), variations in ambient temperature, and physical changes to the standard reference pucks over time (e.g., surface oxidation, moisture uptake, minor contamination). Again, repeated measurement of the same standards helps with identifying such drift.

Errors that are likely to be fixed throughout the whole series of analyses include spectral line overlaps, matrix absorption effects, and secondary fluorescence. Spectral line overlaps are problematic when the fluorescence peak of an abundant element in the sample falls on or near the peak of the element currently being measured, thereby inflating its concentration (e.g., Sr is high concentration in carbonatite-matrix standards and its Kβ will always inflate the lower abundance Zr Kα readings). Matrix absorption (a.k.a. absorption edge effects) occurs when an element that is not being analysed but is abundant in the sample absorbs the incoming X-rays and/or outgoing fluorescence, systematically suppressing the signal (e.g., high-Ca matrices suppress Fe, Co and Zn fluorescence above the Ca K-edge). Secondary fluorescence (or enhancement) occurs when an abundant element's fluorescence excites another element, artificially inflating its measured count rate. The instrument's deconvolution software is supposed to correct for known overlaps and matrix effects, but it is calibrated for typical geological matrices that may not capture more extreme ones, resulting in it potentially systematically overdoing or underdoing the elemental correction for a given standard.

1.2 Reference Standards

Alongside our inventory of real samples, the USGS has provided five geochemical reference materials that are measured several times over the course of a session (nominally near its beginning, middle, and end):

- Carbonate-rich shale
- Marine shale
- Black shale
- Carbonatite
- Phosphate

The five standards span a wide range of matrix compositions and, collectively, a wide dynamic range of concentrations for most elements of interest. Certified reference elemental values and their associated uncertainties are provided by the USGS and are denoted x_j and $\sigma_{x,j}$, respectively, for each standard j (Table 1).

Table 1: Certified elemental concentrations (ppm) and associated 1σ uncertainties ($x_j \pm \sigma_{x,j}$) for five USGS reference materials used for pXRF calibration. Values below the certified detection limit are shown as '—'. Carb. shale = Carbonate-rich shale.

Element	**Carb. shale**	**Marine shale**	**Black shale**	**Carbonatite**	**Phosphate**
			value ± 1σ uncertainty (ppm)		
Major elements					
Mg	26800 ± 1206	15683 ± 60	12300 ± 77	7540 ± 181	6400 ± 51
Al	34500 ± 1111	111132 ± 212	80300 ± 409	1958 ± 212	12000 ± 138
Si	131800 ± 982	222717 ± 400	235700 ± 1217	16222 ± 514	61400 ± 316
P	1400 ± 288	1615 ± 9	400.0 ± 0.1	—	113500 ± 22979
S	15300 ± 1100	7150 ± 80	58400 ± 470	—	2000.0 ± 0.1
K	13800 ± 8301	28640 ± 83	31800 ± 281	1328 ± 166	900 ± 27
Ca	59900 ± 1215	21084 ± 71	11800 ± 201	345200 ± 2716	282600 ± 839
Ti	1500 ± 150	5126 ± 18	3600 ± 47	1500 ± 12	900 ± 34
Fe	21200 ± 1088	67912 ± 155	66400 ± 544	20562 ± 700	39200 ± 199
Trace elements					
V	130.0 ± 6.0	220 ± 1.4	219.1 ± 3.6	110.0 ± 6.0	82.1 ± 2.1
Cr	30.0 ± 3.0	109.0 ± 1.0	98.3 ± 3.6	$5.00 \pm <0.01$	49.2 ± 5.2
Mn	267.0 ± 34.0	1162 ± 8	$200.0 \pm <0.1$	3330 ± 3330	8400 ± 40
Co	12.0 ± 1.5	22.7 ± 0.1	40.6 ± 0.7	$0.50 \pm <0.01$	29.4 ± 2.7
Ni	$29.0 \pm <0.1$	82.8 ± 0.8	192.9 ± 16.5	13.0 ± 1.0	32.0 ± 4.2
Cu	66.0 ± 9.0	31.0 ± 0.6	176.7 ± 3.2	$5.00 \pm <0.01$	23.0 ± 5.0
Zn	74.0 ± 9.0	186.0 ± 1.7	62.5 ± 7.8	87.0 ± 1.5	26.4 ± 3.7
As	67.0 ± 5.0	25.7 ± 0.7	46.2 ± 0.8	—	24.8 ± 0.9
Se	$4.00 \pm <0.01$	—	6.31 ± 0.87	—	5.38 ± 9.63
Rb	—	147.0 ± 1.0	140.0 ± 5.0	—	5.08 ± 2.03
Sr	420.0 ± 30.0	178.0 ± 1.4	86.7 ± 1.7	12000 ± 50	703.6 ± 3.1
Y	$13.0 \pm <0.1$	36.5 ± 0.1	26.7 ± 0.6	81.0 ± 0.5	631.0 ± 3.9
Zr	$53.0 \pm <0.1$	134.0 ± 1.6	144.2 ± 2.3	65.0 ± 3.0	58.7 ± 10.8
Nb	—	15.3 ± 0.1	8.81 ± 0.10	3900 ± 60	7.67 ± 1.51

Continued on next page

Table 1 continued

Element	**Carb. shale**	**Marine shale**	**Black shale**	**Carbonatite**	**Phosphate**
	value ± *1σ uncertainty (ppm)*				
Mo	35.0 ± 0.9	2.40 ± 0.10	159.4 ± 4.3	—	—
Ag	—	—	—	—	—
Cd	—	0.10 ± 0.10	0.10 ± 0.10	—	—
Sn	$2.00 \pm <0.01$	3.30 ± 0.10	—	—	—
Sb	3.00 ± 0.50	1.01 ± 0.10	4.51 ± 0.10	—	3.29 ± 0.10
Ba	290.0 ± 40.0	788.0 ± 7.7	454.1 ± 29.6	1000 ± 50	219.7 ± 1.7
La	20.0 ± 1.8	52.5 ± 0.6	33.5 ± 0.5	750.0 ± 10.0	495.8 ± 4.9
Ce	36.0 ± 4.0	108.0 ± 0.9	64.6 ± 1.1	1700 ± 100	928.8 ± 10.2
Pr	—	12.6 ± 0.1	8.23 ± 0.10	150.0 ± 6.0	96.7 ± 1.2
Nd	16.0 ± 1.7	49.2 ± 0.5	31.6 ± 1.2	480.0 ± 15.0	351.0 ± 9.9
W	3.00 ± 0.06	1.60 ± 0.10	—	—	—
Hg	—	—	—	—	—
Pb	38.0 ± 4.0	35.0 ± 0.1	52.2 ± 0.9	—	76.7 ± 4.1
Bi	—	0.70 ± 0.10	0.10 ± 0.10	—	—
Th	5.00 ± 0.21	15.8 ± 0.1	9.31 ± 0.10	10.0 ± 1.0	10.7 ± 0.7
U	5.00 ± 0.40	5.76 ± 0.10	25.4 ± 0.5	11.0 ± 0.6	30.5 ± 0.7

1.3 Standard selection and initial elemental screening

Not every standard is suitable for calibrating every element and individual standards have been excluded on a per-element basis prior to fitting for one of five reasons:

1. Since we will be fitting linear relationships that are forced to pass through zero, those standards with certified concentration of zero (or indistinguishable from zero at the level of the reference uncertainty) carry no information about the calibration slope and are automatically excluded. An example of this would be Se in marine shale.

2. Similarly, due to matrix absorption and interference effects, some elements that are truly present in standards never return a pXRF concentration above its limit of detection and are also automatically excluded. Examples of this would be Co and K in phosphate (suppressed by high Ca matrix), V in phosphate (severe La and Ce interference), and Th in carbonatite (suppressed by high Ca/REE matrix).

3. Certain standards have extreme enrichment in a specific element that are not representative of the true samples (i.e., they are only a minor or trace component of regular shales and carbonates). Including such a standard extends the calibration line far beyond the concentration range of the true samples and introduces potential non-linearity into the calibration curve due to detector dead-time or matrix absorption effects. They therefore risk a single high-leverage point dominating the regression. Example elements that are extremely enriched in the carbonatite are Ca ($\sim 345,000$ ppm), Sr ($\sim 12,000$ ppm), Ce (~ 1700 ppm) and La (~ 750 ppm), and in the phosphate are Ca ($\sim 280,000$ ppm), P ($\sim 113,500$ ppm), Mn (~ 8400 ppm) and Ce (~ 930 ppm). Any time the phosphate or carbonatite standard has a concentration greater than four times the most enriched of the three shale standards, it is therefore discarded from the calibration.

4. Certain standards that are extremely enriched in one of the above elements can have a knock-on effect in other elements due to the aforementioned spectral interference and matrix absorption effects. Identifying these issues is what has taken me the longest and requires documenting in more detail.

5. The final suite of accepted standards should fall along an approximately linear array for our calibration model (i.e., a linear fit through zero) to be suitable. If the remaining standards for a given element still does not have a session-pooled coefficient-of-variation (see below) smaller than 0.5 after application of rules 1–4, that element is rejected as unusable. Note that this property can only be assessed when there is more than one

standard remaining, so if at any point in the screening process the number of standards remaining for a given element in a given session falls below two, it is automatically excluded.

After excluding standards for some elements based on the first three rules above and discarding any remaining elements with fewer than two usable standards, we next assess whether the remaining standards form a coherent quasi-linear array that passes roughly through the origin. To do so, we compute the coefficient-of-variation (CV) of the ratio $\bar{y}_j/x_j$ using

$$\mathrm{CV} = \frac{1}{\bar{r}}\sqrt{\frac{1}{n-1}\sum_{j=1}^{n}(r_j-\bar{r})^2}, \qquad r_j = \frac{\bar{y}_j}{x_j}, \quad \bar{r} = \frac{1}{n}\sum_{j=1}^{n} r_j \tag{1}$$

where n is the number of standards retained for that element and $\bar{y}_j$ is the session-pooled mean pXRF concentration for each of those standards. We then visually inspect the trends from specific sessions, using elevated CV as a guide, and find that one or sometimes two of the phosphate and carbonatite standards systematically do not sit on the same linear array as the others despite both having a true composition that is intermediate to the other standards and lying within the realistic range of the real samples (Figure 1). This behaviour is caused by the aforementioned interference and absorption effects and is handled by a manual exclusion procedure.

A good example of a problematic element would be Zr in carbonatite, which sits substantially above the 1:1 line compared to the other four standards (Figure 1). The explanation for this behaviour is that the carbonatite is extremely enriched in Sr ($\sim 12,000$ ppm) compared to standard sediments like shales (< 500 ppm). The Sr Kβ occurs at 15.83 keV, while Zr Kα sits at 15.77 keV, which is a difference of only 0.06 keV, and so Sr is producing an enormous positive interference on Zr for the carbonatite. Similarly, pXRF V counts in both the carbonatite and phosphate are systematically erroneous (usually undetected). Carbonatite and phosphate both have high contents of La and Ce and, since La Lβ (5.04 keV) and Ce Lα (4.84 keV) sit right on top of the much less abundant V Kα (4.95 keV), their signal swamps the V concentration estimate. As a final example, Ti was noticeably too low for both the carbonatite and phosphate, which are both very rich in Ca. Ca has a K-absorption edge at 4.04 keV, while Ti Kα at 4.51 keV sits just above it, meaning Ca efficiently absorbs outgoing Ti fluorescence photons on their way to the detector. These examples and others were handled by manually excluding the affected standard(s) from that specific elemental calibration.

Recomputing session-pooled CVs after these manual exclusions demonstrates a substantial improvement in linearity for some elements. For example, exclusion of carbonatite drops CV for Zr from 0.179 to 0.080, V from 0.056 to 0.035, K from 0.480 to 0.036, and Mg from 0.245 to 0.048. Exclusion of phosphate and carbonatite drops CV for Ti from 0.379 to 0.016 and Cr from 1.534 to 0.073. As part of the final screening step (rule 5), we only accept elements that have $CV < 0.5$, which results in only Sb being discarded as it has four standards but an incoherent array (with marine shale a significant outlier). To summarise the end result of these various screening exclusions:

- No coherent calibration array exists for Ag, Bi, Cd, Hg, Sb, Se, Sn, and W.
- Carbonatite and phosphate are both excluded from calibration of Ca, Ce, Co, Cr, Cu, K, La, Mn, Mo, Nb, P, S, Th, Ti, U and V.
- Carbonatite only is further excluded from calibration of As, Ba, Mg, Pb, Rb, Sr, Zn, and Zr.
- Phosphate only is further excluded from calibration of Ni and Y.

This pattern of elements that are manually excluded due to rule 4 is more or less consistent with interference physics. Elements whose fluorescence energies are affected by the Ca K-edge absorption (Ti, Co, Zn, Cu, Ba, Zr) or REE L-line overlap (V, Ba) need carbonatite excluded, while elements affected by the extreme P/Ca matrix of Phosphate rock (Ni, Co, Y, La, Ce) or the U/Th L-line cross-contamination (Th) need phosphate excluded. The elements needing neither exclusion (Al, Si and Fe) are those where both of the non-shale standards happen to plot on the same calibration line as the shales due to one of: (i) negligible interference at those energies; (ii)

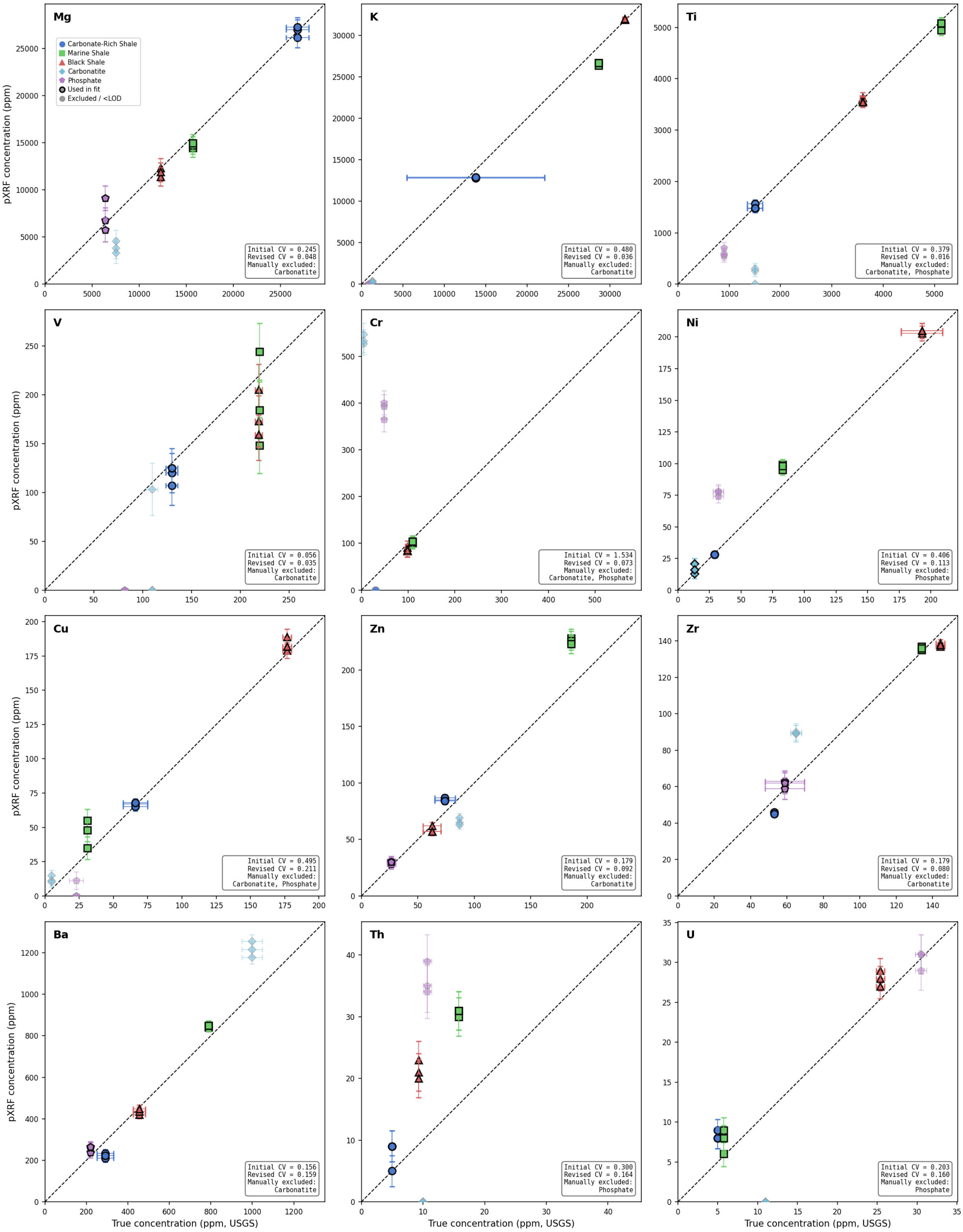

Figure 1: **Elemental screening requiring manual exclusion of standards.** Data is from a session on 27^{th} May 2025. Each panel shows all standard measurements made that remain after application of screening rules 1–3. Pale symbols are standards that are subsequently manually removed. The coefficient-of-variation (CV) across all sessions is reported both before and after standard removal. Note that, in some cases, plotted 1σ errors are smaller than the symbol size. Only elements that required manual standard exclusions are shown.

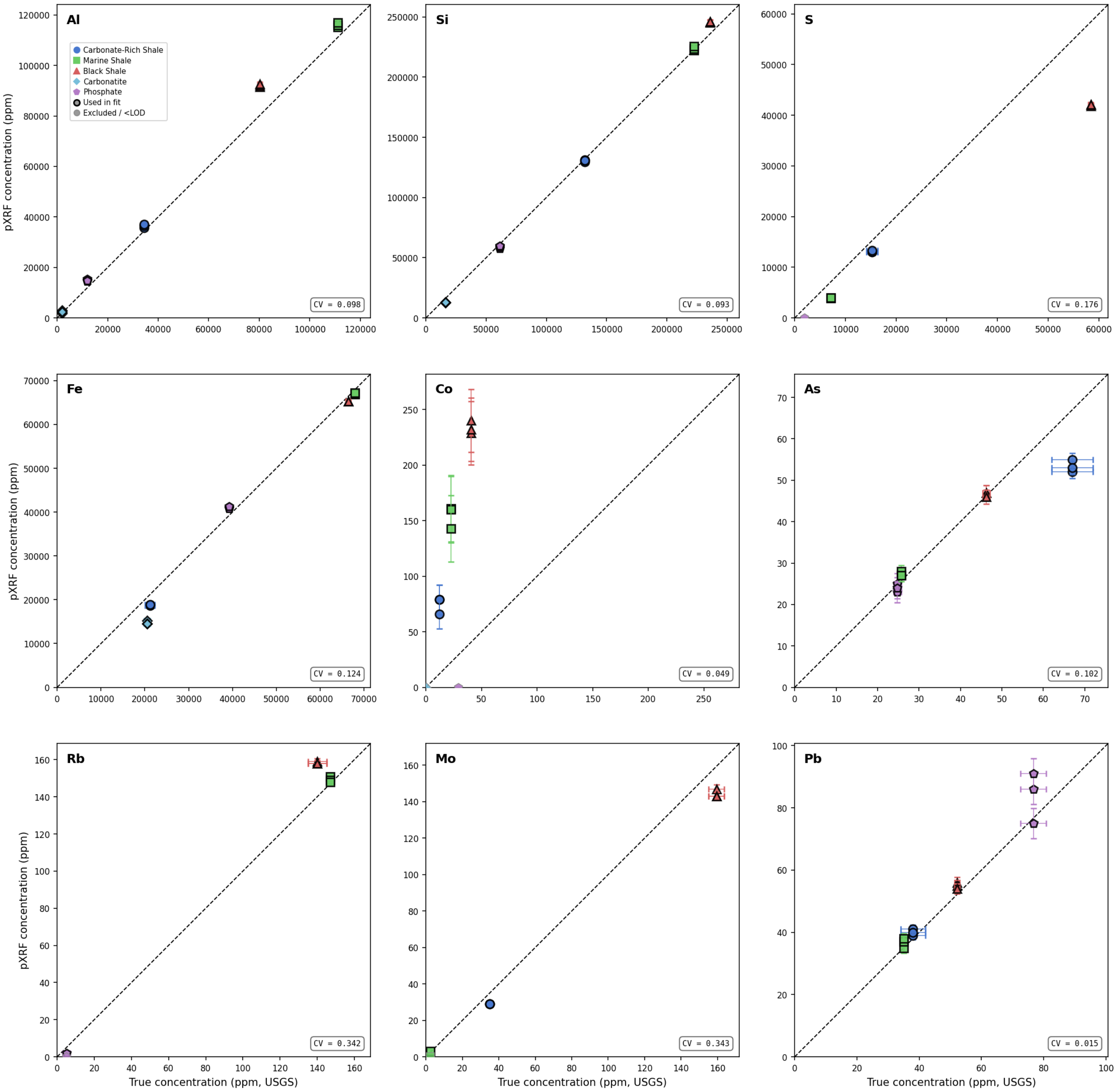

Figure 2: **Elemental screening requiring no further manual exclusion of standards.** Data is from a session on 27^{th} May 2025. Each panel shows all standard measurements made that remain after application of screening rules 1–3 and their associated coefficient-of-variation (CV) across all sessions. Note that, in some cases, plotted 1σ errors are smaller than the symbol size. Only elements that did not require manual standard exclusions are shown.

the absorption and enhancement effects approximately cancelling out; or (iii) the detector is doing a good job of internal correction.

1.4 Formal model calibration

We assume a linear relationship between the true elemental concentration, x, and the raw pXRF-reported concentration, y, that is forced through the origin according to

$$y = m\,x \tag{2}$$

where m is the calibration slope. The intercept is physically constrained to be zero based on the argument that a sample with no detectable level of an element should return a zero instrument response in the absence of spectral interference. Elements affected by significant and systematic non-zero offsets are generally excluded by the various screening criteria above.

Both the reference values x_j and the pXRF measurements y_{ji} (where j indexes the standard and i indexes each replicate measurement of that standard within the session) carry uncertainties, so ordinary least-squares regression (which assumes errors only in y) is inappropriate. We instead employ an errors-in-variables (EIV) approach, which is equivalent to Deming regression constrained to pass through the origin (e.g. Carroll *et al.*, 2006). For a given element and session, each individual replicate measurement i of standard j contributes a data point $(x_j,\, \sigma_{x,j},\, y_{ji},\, \sigma_{y,ji})$, where x_j is the certified reference concentration (constant across all replicates of the same standard), $\sigma_{x,j}$ is the certified reference uncertainty, y_{ji} is the individual pXRF reading, and $\sigma_{y,ji}$ is the effective pXRF uncertainty. The EIV chi-squared objective function is

$$\chi^2(m) = \sum_{j,i} \left[\frac{(x_j - X_{ji})^2}{\sigma^2_{x,j}} + \frac{(y_{ji} - mX_{ji})^2}{\sigma^2_{y,ji}} \right] \tag{3}$$

where X_{ji} is the latent true concentration for replicate i of standard j. For fixed m, the optimal X_{ji} has the closed-form solution

$$X_{ji}(m) = \left(\frac{x_j}{\sigma^2_{x,j}} + \frac{m\,y_{ji}}{\sigma^2_{y,ji}} \right) \Big/ \left(\frac{1}{\sigma^2_{x,j}} + \frac{m^2}{\sigma^2_{y,ji}} \right) \tag{4}$$

which reduces the problem to a one-dimensional minimisation in m. The effective pXRF uncertainty $\sigma_{y,ji}$ is a random error that has two distinct components. First, a 1σ counting uncertainty, $\sigma^c_{y,ji}$, is reported by the instrument itself. Unfortunately, this factor substantially underestimates the true measurement-to-measurement variability of standard readings. Repeated measurements of the same standard within a single analytical session differ not only due to counting statistics but also due to slight variations in sample positioning geometry on the analyser window, surface heterogeneity of the pressed-powder reference puck, and short-timescale ambient temperature fluctuations. To account for these additional sources of scatter, an empirical intrinsic scatter term, ε_j, is introduced.

To estimate ε_j, we work session by session (each session is indexed by k). Within a session, the standards are measured in several discrete blocks spread over its duration (typically three, located near the beginning, middle, and end of the session). A session contributes to the estimate only if it contains at least two such blocks, ensuring that the scatter reflects genuine re-positioning and short-term drift across the session rather than a single tightly-clustered burst of consecutive readings. For each qualifying session, the mean, $\bar{y}_j^{(k)}$, and unbiased sample variance, $s^2_{j,k}$, are then computed across all $n_{j,k}$ individual readings of each standard, j, in that session (pooled across its blocks), using

$$\bar{y}_j^{(k)} = \frac{1}{n_{j,k}} \sum_i y_{ji}^{(k)}, \quad s^2_{j,k} = \frac{1}{n_{j,k}-1} \sum_{i=1}^{n_{j,k}} \left(y_{ji}^{(k)} - \bar{y}_j^{(k)} \right)^2 \tag{5}$$

The intrinsic scatter is then estimated as the pooled within-session standard deviation across all K qualifying sessions according to

$$\varepsilon_j = \sqrt{ \sum_{k=1}^{K} (n_{j,k}-1)\, s^2_{j,k} \Big/ \sum_{k=1}^{K} (n_{j,k}-1) } \tag{6}$$

which is the standard pooled-variance estimator. This formulation weights each session's contribution by its degrees of freedom and is therefore robust to the varying number of readings per session (typically 2–3 per block across 2–3 blocks, giving $n_{j,k} \sim 4$–9). By construction, ε_j is estimated from the *scatter* of readings around their session mean rather than from the mean values themselves, so it is independent of longer-term session-to-session drift in the calibration slope m.

The effective measurement uncertainty used in the EIV fit then becomes

$$\sigma_{y,ji} = \sqrt{\left(\sigma^c_{y,ji}\right)^2 + \varepsilon_j^2} \tag{7}$$

which adds the intrinsic scatter in quadrature with the counting uncertainty. The approach effectively down-weights standards whose readings are poorly reproducible due to positioning or surface effects (e.g., Ca in the marine shale, where $\varepsilon \approx 390$ ppm compared to typical reported counting errors of $\sigma_y \sim 50$–120 ppm).

Equations (3) and (4) are numerically minimised using a bounded scalar optimiser (SciPy `minimize_scalar` with method `bounded`), starting from the initial regular weighted-least-squares estimate

$$m_0 = \sum_{j,i} \frac{x_j\, y_{ji}}{\sigma^2_{y,ji}} \Big/ \sum_{j,i} \frac{x_j^2}{\sigma^2_{y,ji}} \tag{8}$$

and raw uncertainty on the best-fit slope is estimated from the curvature of $\chi^2(m)$ around the minimum

$$\sigma_m^{\mathrm{raw}} = \sqrt{2\left(\frac{\partial^2 \chi^2}{\partial m^2}\right)^{-1}} \tag{9}$$

which is computed using a three-point finite-difference approximation. The reduced chi-squared is also computed using

$$\chi^2_\nu = \frac{1}{\sum_j n_{j,k} - 1}\chi^2_{\min} \tag{10}$$

and is found to be close to one for many elements (e.g., $\chi^2_\nu \approx 0.3$–1.0 for Ti, Mn, V, Pb, and Cr), confirming that the intrinsic scatter term, ε_j, is capturing the dominant source of excess variability in pXRF data for those elements. However, several high-concentration elements retain $\chi^2_\nu \gg 1$ (e.g., $\chi^2_\nu \approx 27$ for Fe, ≈ 14 for Si and K), indicating that there remain additional unmodelled sources of scatter. Possible concentration-dependent explanations include slight non-linearity in detector response at high count rates or matrix-dependent self-absorption that have not been identified during the manual removal of potentially anomalous standards. To obtain more realistic slope uncertainties, we therefore apply the rescaling factor

$$\sigma_m = \sigma_m^{\mathrm{raw}} \times \sqrt{\chi^2_\nu} \tag{11}$$

when $\chi^2_\nu > 1$. This rescaling is equivalent to allowing effective measurement uncertainty to be determined by the observed scatter rather than the nominal counting error in cases when the former dominates the latter, which is standard practice in weighted regression when error estimates are believed to be underestimates (Bevington & Robinson, 2003). After calibration, corrected pXRF-derived concentrations can therefore be calculated as $x_{\mathrm{corr}} = y/m$ and their uncertainty, $\sigma_{x_{\mathrm{corr}}}$, propagates as

$$\frac{\sigma_{x_{\mathrm{corr}}}}{x_{\mathrm{corr}}} = \sqrt{\left(\frac{\sigma^c_y}{y}\right)^2 + \left(\frac{\sigma_m}{m}\right)^2} \tag{12}$$

where it is important to note that only the instrument-reported counting uncertainty, σ^c_y, is present. The intrinsic scatter term, ε_j, does not enter this equation since it is specific to each reference standard and cannot be meaningfully transferred to unknown samples of different matrix and surface characteristics. However, its effect on calibration accuracy is already propagated through σ_m. An example of this full workflow applied to data collected on 20^{th} May 2025 is shown in Figure 3.

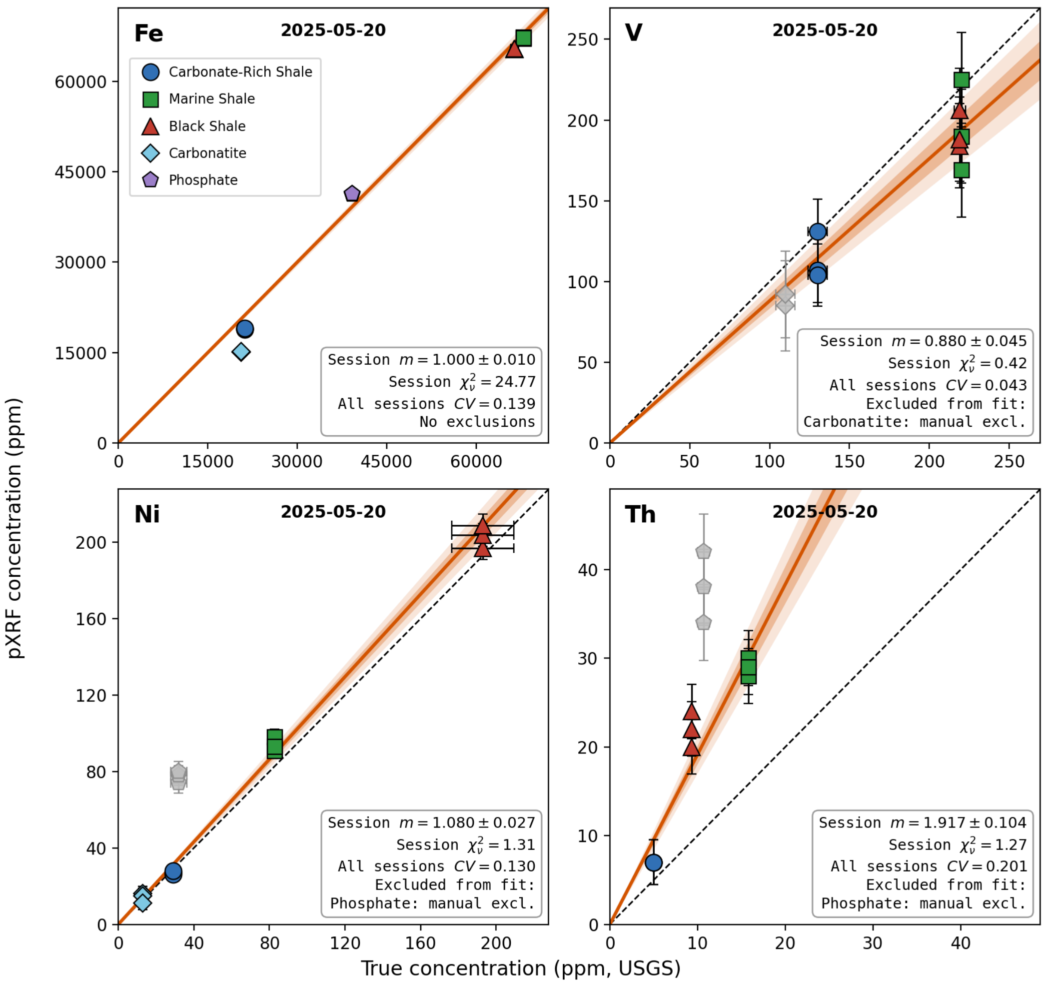

Figure 3: **Example errors-in-variables regression results.** Panels show all standard measurements made in a session on 20^{th} May 2025 for Fe, V, Ni and Th. Manually excluded symbols from rule 4 of the screening are shown in grey. While the V for carbonatite looks reasonable in this session, across the whole set of sessions, it is highly variable due to the aforementioned interference caused by high Ce and La concentrations. Note the elevated reduced chi-squared value for Fe, indicating that there are likely still unmodelled interference or matrix absorption effects present.

2 Calibration results

2.1 Session consistency

The regression is performed independently across each element for each analytical session (defined by calendar date). To be included, a session requires at least two distinct standards and must contain at least four individual replicated measurements (i.e., since we are fitting one parameter, m, we therefore keep at least three degrees of freedom for the χ^2_ν computation). This criteria results in Nd and Pr becoming uncalibratable. To assess whether a single mean calibration factor can be used consistently across all sessions, or whether session-specific factors are required, we compute a precision-weighted mean slope

$$\bar{m} = \sum_k^N \frac{m_k}{\sigma^2_{m,k}} \Bigg/ \sum_k \frac{1}{\sigma^2_{m,k}} \tag{13}$$

where index k refers to each specific session across the total of N sessions. We can then test for between-session homogeneity using

$$\chi^2_{\text{hom}} = \sum_k^N \frac{(m_k - \bar{m})^2}{\sigma^2_{m,k}} \tag{14}$$

which follows a χ^2 distribution with $\upsilon = N - 1$ degrees of freedom under the null hypothesis that all sessions share a common calibration slope. A reduced homogeneity statistic, $\frac{1}{N-1}\chi^2_{\text{hom}}$ is used as a criterion for assessing

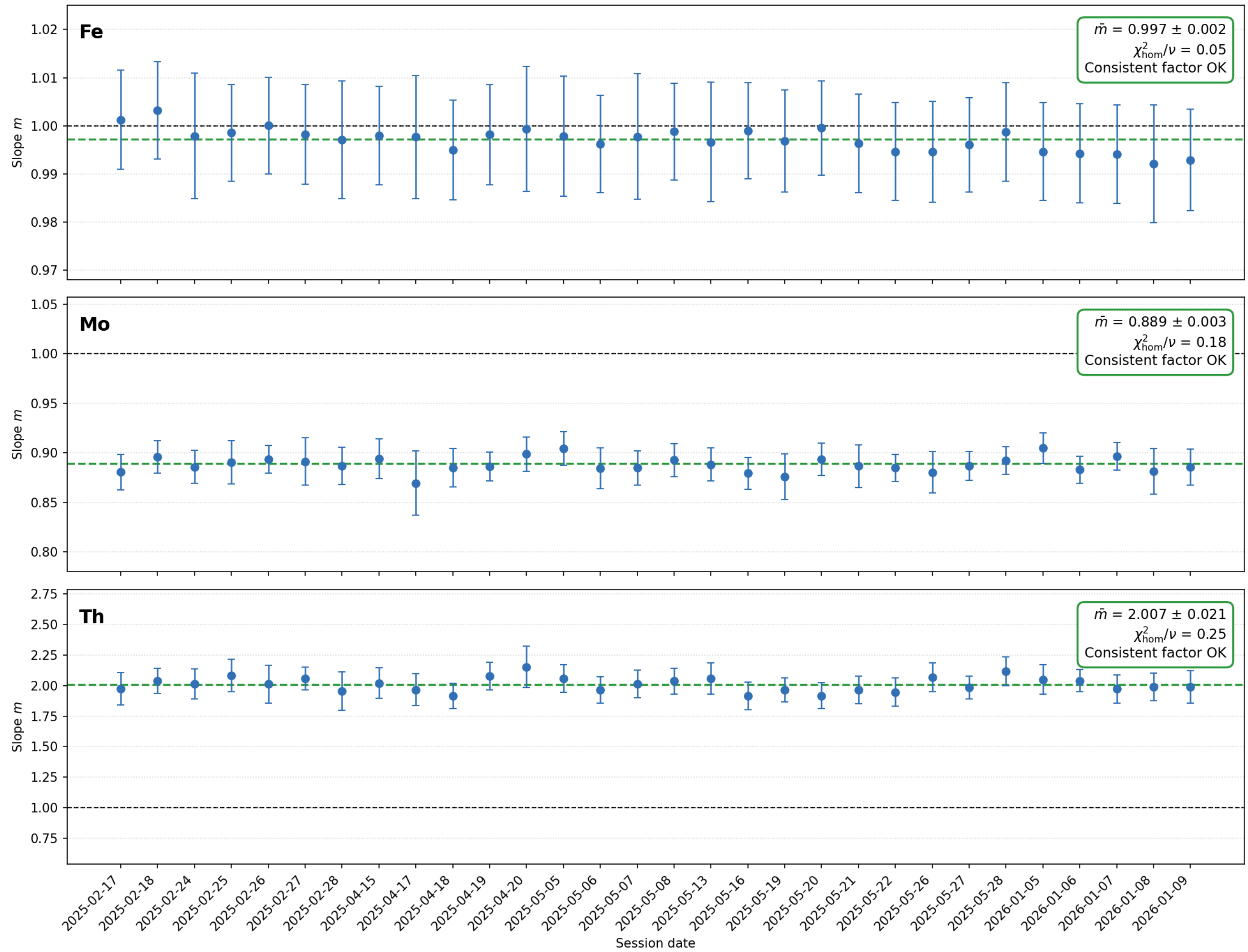

Figure 4: **Example session-by-session regression results.** $\bar{m}$ is the mean calibration factor with 1σ errors and $\chi^2_{\mathrm{hom}}/\upsilon$ is the reduced homogeneity statistic, which should be around one or less for consistent values between sessions. Fe, Mo and Th all exhibit consistent factors within error, such that a single calibration parameter can be applied across all sessions.

the suitability of a single consistent factor. Values substantially larger than one therefore indicate meaningful session-to-session variation and probably require session-specific calibration factors.

For the majority of elements, there is strong session-to-session consistency in the optimal calibration factor. Figure 4 shows results for three such examples. Fe is a major element and has a mean calibration parameter across all sessions of $\bar{m} = 0.997 \pm 0.002$, which is close to the ideal 1:1 line. Mo is a minor element that is consistently under-predicted across the sessions with $\bar{m} = 0.889 \pm 0.003$, while Th is a minor element that is consistently over-predicted with $\bar{m} = 2.007 \pm 0.021$. The majority of trace elements fall within this category, with some particularly relevant examples shown in Figure 5.

However, if you look closely at Fe in Figure 4, you will notice that it actually exhibits a fairly steady drift across all sessions, starting generally slightly higher than the mean and ending up generally slightly lower. In this case, this drift is well within error of the individual calibration slopes and so using a consistent factor is probably still justified. However, the same drift is also visible in several of the other major elements. For K, S, and Ba, this drift is small in comparison to the within-session uncertainty, but it is more significant for Al and Mg (Figure 6). Finally, for Ca, Si, P, and Sr, this drift is no longer continuous, but seems to systematically vary to-and-fro over the course of the sessions (Figure 7). Sr is still homogeneous within slope uncertainties, so a single factor should suffice. Si varies between 0.98–1.05 and Ca varies 0.92–1.02, so the error from using a constant value would be on the order of ±5%, which might be acceptable? But P varies hugely, ranging 0.61–1.23. Moreover its changes are significantly larger than the error on a slope within individual sessions.

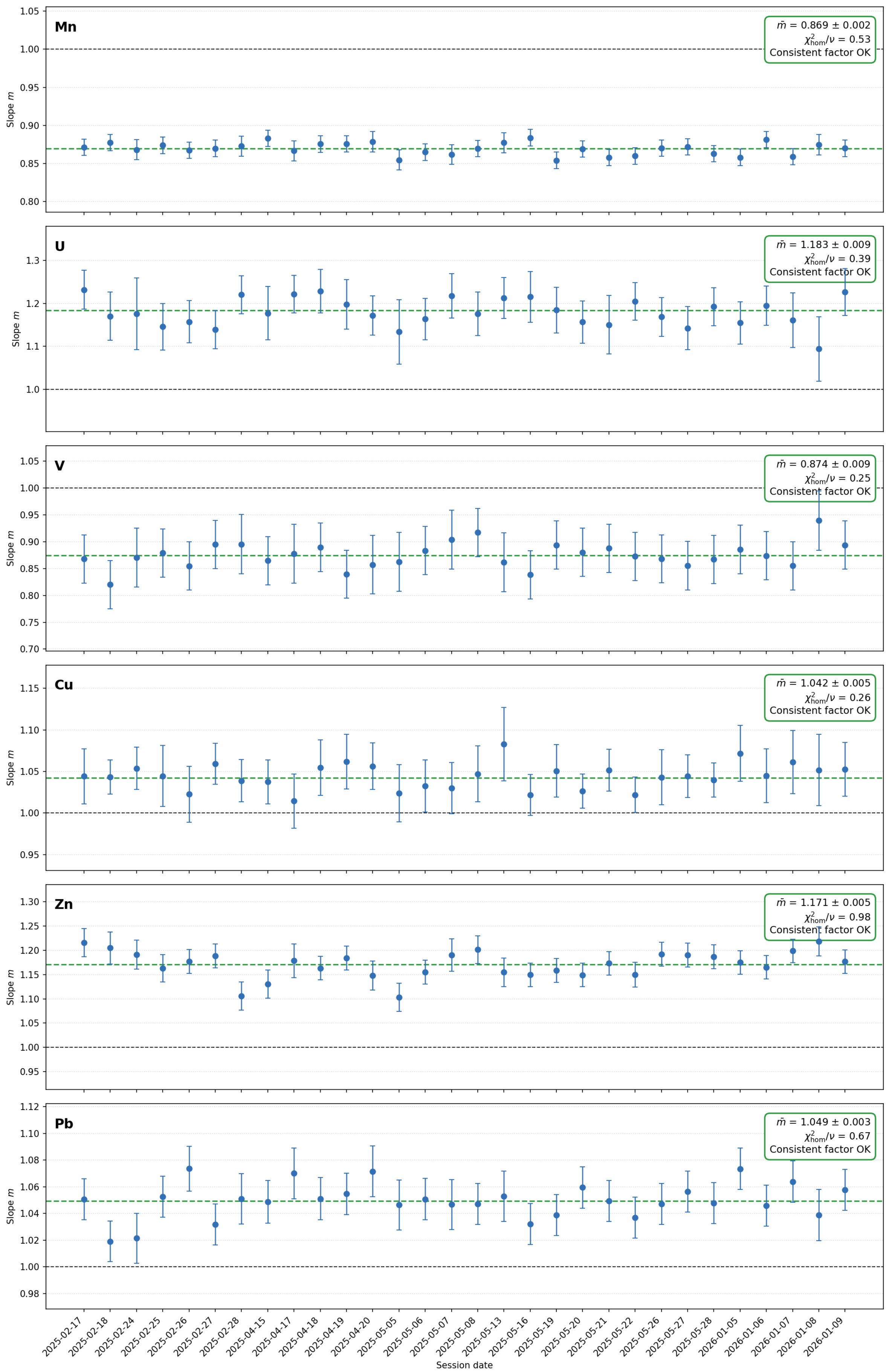

Figure 5: **Session-by-session regression results for relevant trace elements.** $\bar{m}$ is the mean calibration factor with 1σ errors and $\chi^2_{\mathrm{hom}}/\upsilon$ is the reduced homogeneity statistic, which should be around one or less for consistent values between sessions. Mn, U, V, Cu, Zn, and Pb all exhibit consistent factors within error, such that a single calibration parameter can be applied across all sessions, although Zn is borderline.

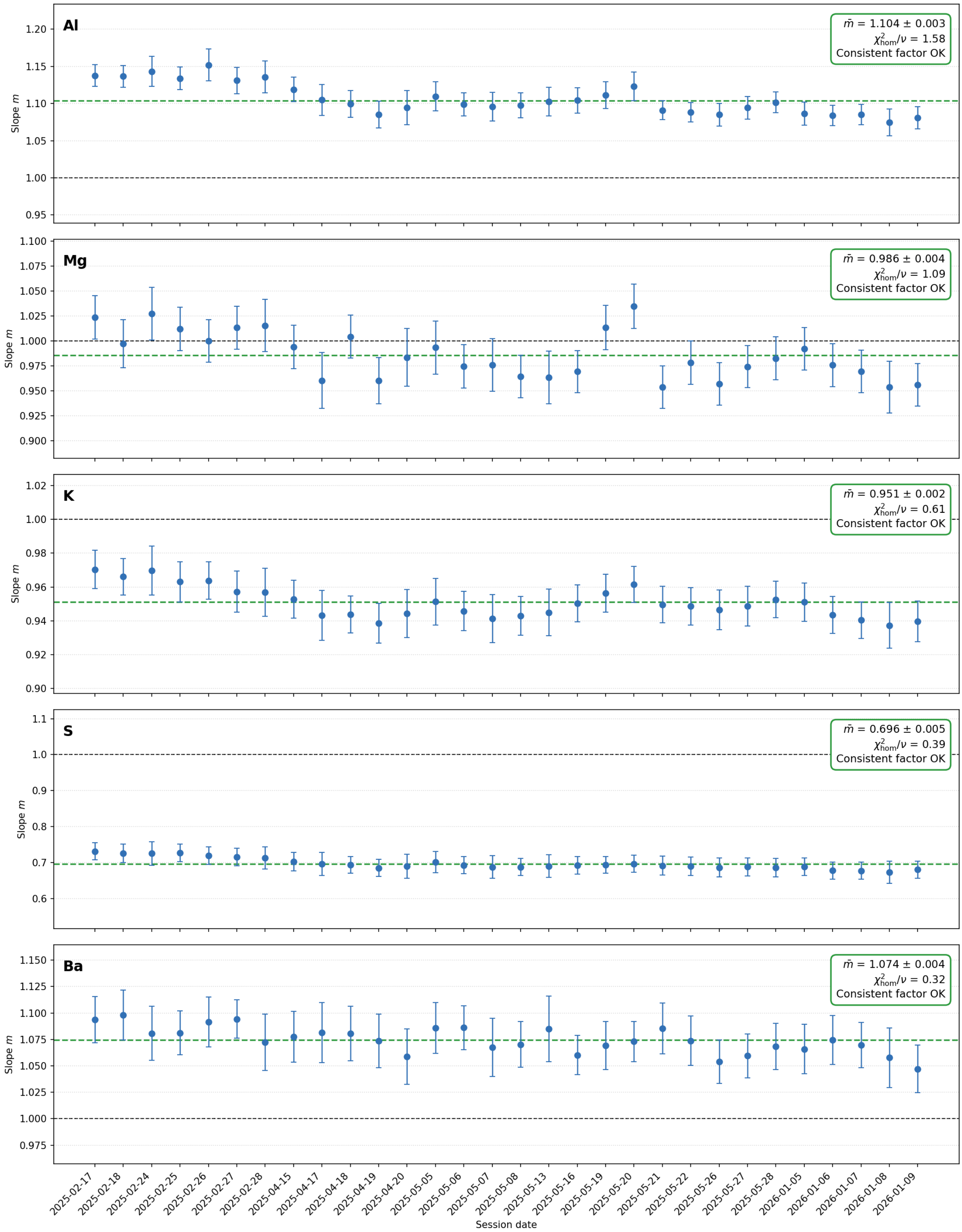

Figure 6: **Session-by-session regression results for some elements exhibiting drift.** $\bar{m}$ is the mean calibration factor with 1σ errors and $\chi^2_{\text{hom}}/\upsilon$ is the reduced homogeneity statistic, which should be around one or less for consistent values between sessions. Al, Mg, K, S, and Ba all drift, but not substantially more than the uncertainties in their individual session-by-session calibration factors.

2.2 Some thoughts on P

A severe anomaly in P occurs between 15^{th} April and 13^{th} May 2025, during which the slope rises sharply from around 0.75 to values greater than 1.20 (Figure 8). Notably, the within-session uncertainty in m also increases by a factor of three from ± 0.04 to ± 0.15, showing that these sessions were consistently less stable than those before and after the anomaly (i.e., the large σ_m values reflect genuine incoherence of the calibration array).

Interestingly, the carbonatite reference material has a certified P concentration of zero (or below detection).

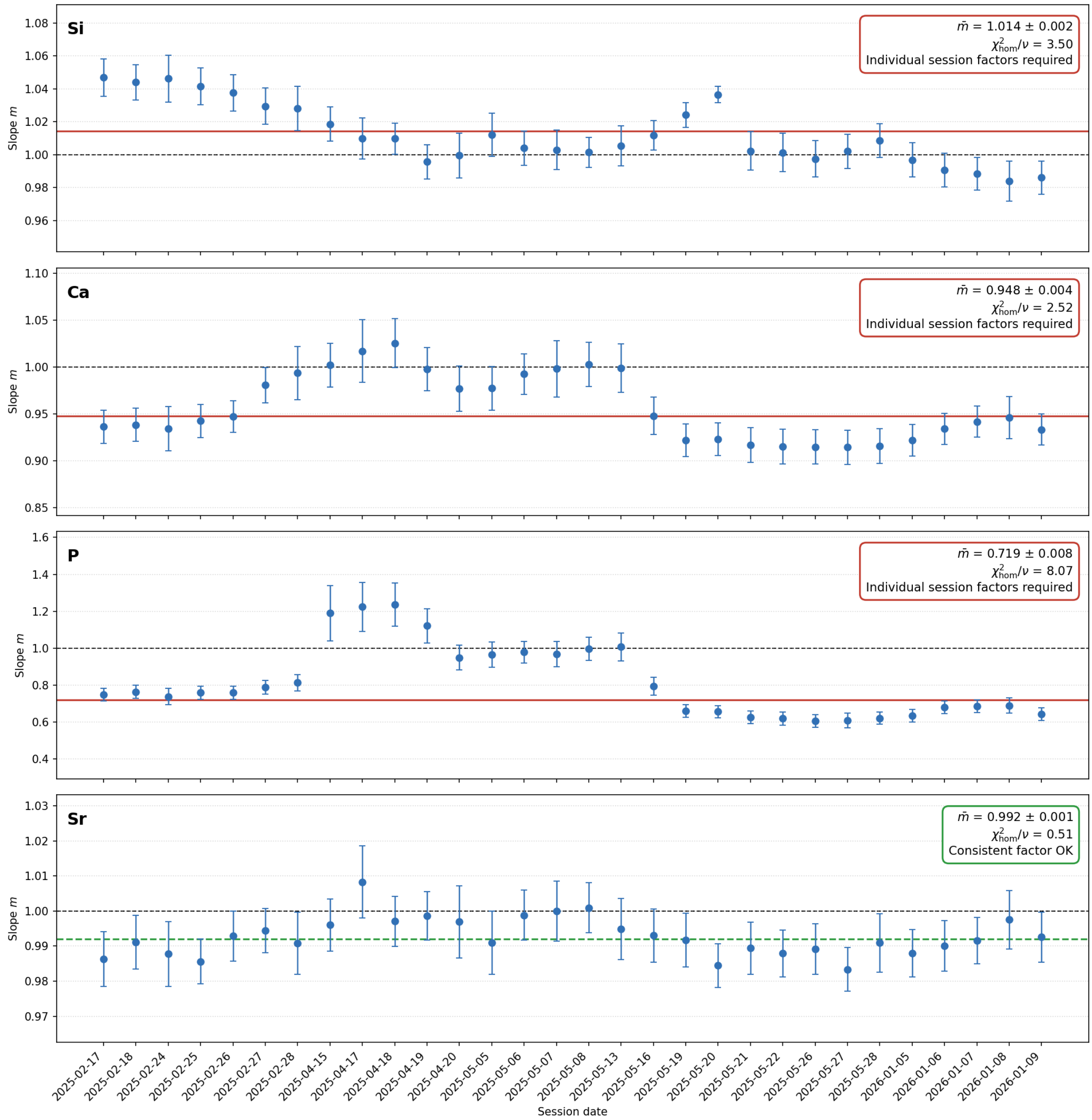

Figure 7: **Session-by-session regression results for some elements exhibiting variable drift.** $\bar{m}$ is the mean calibration factor with 1σ errors and $\chi^2_{\mathrm{hom}}/\upsilon$ is the reduced homogeneity statistic, which should be around one or less for consistent values between sessions. Si, Ca, and P are all more variable than their within-session uncertainty and probably need to be treated with session-specific calibration factors.

As mentioned above, however, the pXRF consistently returns a P reading of approximately 4500–4900 ppm for this standard, which must be caused by spectral interference by one or more of the elements that are abundant in carbonatite (Nb, Sr, La, Ce, Ca, Mn), whose fluorescence lines partially overlap the P Kα emission. Crucially, the spurious pXRF P signal measured on the carbonatite standard tracks the same temporal anomaly as the calibration slopes derived from the other three standards, also rising ~300–400 ppm above the baseline and exhibiting greatly increased replicate scatter. Since the carbonatite composition is fixed, this variation must originate from the instrument. Given that P Kα emission occurs at ~ 2.0 keV and Ca Kα at 3.7 keV, it seems that the instrumental instability is primarily affecting the low-energy portion of the fluorescence spectrum.

Finally, I looked into what was happening to P in individual specific sessions and found that, in some cases, there appears to be systematic drift throughout the course of the session. There were only two or three bad cases, and I also confirmed that it is not occurring for any of the other elements. I therefore think we can get away with using individual session calibration factors for the majority of sessions.

All of the associated python code base can be found at `https://github.com/mjh217/pXRF-calibrate`

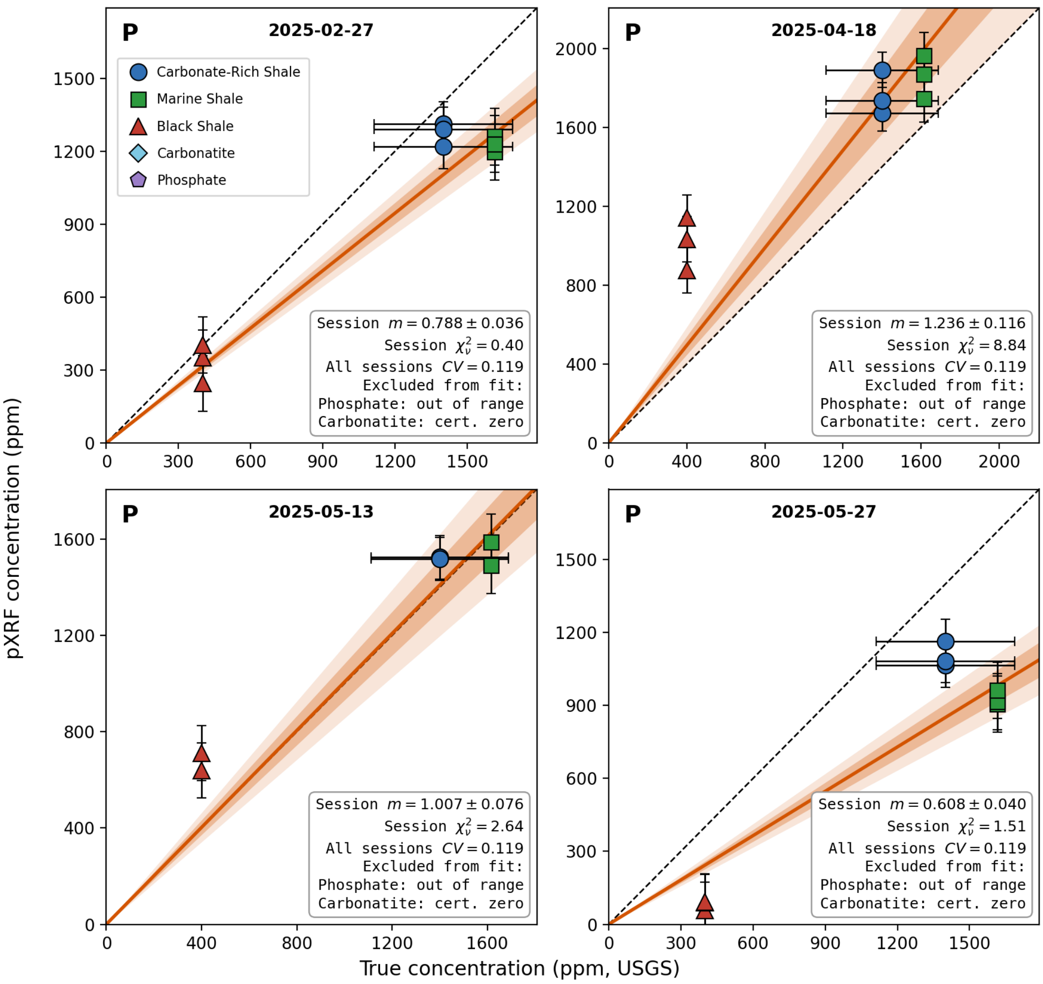

Figure 8: **Errors-in-variables regression results for P during four different sessions.** Panels show all standard measurements made in sessions on 27^{th} February, 18^{th} April, 13^{th} May and 27^{th} May 2025. Note the highly variable calibration results.

References

Bevington, P. R. & D. K. Robinson (2003). *Data reduction and error analysis for the physical sciences.* McGraw-Hill, New York, 3 edn.

Carroll, R. J., D. Ruppert, L. A. Stefanski, & C. M. Crainiceanu (2006). *Measurement error in nonlinear models: A modern perspective.* Taylor and Francis, Boca Raton, Florida, 2 edn.